

The Wrong Pill Should Never Go Unnoticed

A medication-safety assistant for seniors and Canadians with language barriers — it reads the prescription, verifies pills by photo, and answers questions in your language. When it isn't sure, it says so.

CAPSTONE MVP · DECISION-SUPPORT ONLY

The Question

The riskiest moment in home medication use is mundane: **a loose pill out of its bottle**. In a weekly organizer or a shared household, many tablets are small, white and round — and "which pill is this?" is a daily question with a non-trivial cost of getting it wrong.

1 in 4

Canadian seniors is prescribed 10+ drug classes (CIHI)

~5×

adverse-drug-reaction hospitalization risk at 10+ medications (CIHI)

Verify, Don't Identify

- Open-set pill ID from consumer photos is unreliable — the NLM challenge winner reached only **43% top-5**.
- Our measured ceiling on the 7,055 marketed Canadian tablets & capsules: **38.4%** with a perfect imprint read, **1.3%** without.
- Canadian pills genuinely collide: **13 different products** are all the same "blue diamond tablet, SIL 25".
- Within one patient's 5-medication profile, collision odds fall to **~0.3%** — so MyPillSafe verifies against **your list**, with an explicit right to abstain, and never guesses against the whole formulary.

Objectives

- Turn a prescription photo into a DIN-linked medication profile the **user approves line by line**.
- Warn when a photographed pill **does not match** the profile — tuned so a false "verified" is the rarest event.
- Answer medication questions **only** from Health Canada monographs, cited, in the user's language.
- Abstain honestly whenever the evidence is thin — **abstention is a designed outcome, not a failure**.

Technology

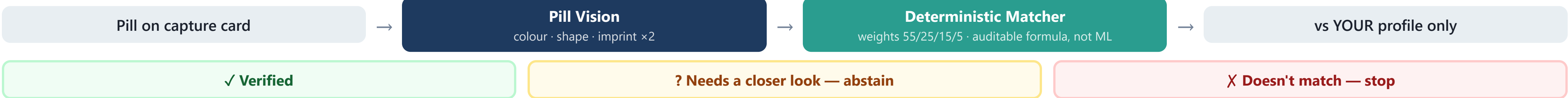
FRONTEND	React 18 · TypeScript · PWA · EN/FR i18n (756 keys, lockstep-guarded)
BACKEND	FastAPI · PostgreSQL · 227 automated tests
VISION	FastSAM (zero-shot) · ResNet18 shape · colorimetric colour · PP-OCR dual-read imprint
LANGUAGE	qwen2.5:7b local proposer + 6 deterministic guardrails · Claude Haiku 4.5 cloud voice
DATA	Health Canada DPD · 6,803 product monographs · 3.9 M passages
OPS	Docker · nginx/TLS · GPU brains sidecar over Tailscale

The Five-Brain Architecture — the AI that talks is never the AI that decides

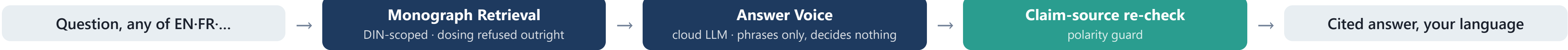
1 · PRESCRIPTION → PROFILE



2 · PILL PHOTO → VERDICT



3 · QUESTION → CITED ANSWER



Why the split is deterministic-first: in evaluation, a language model answered a sulfa-allergy question against its own cited contraindication source. Every decision on this poster's flows is made by an auditable rule; the only generative step is phrasing — and it is re-checked against its sources.

Measured Results — reported honestly

180/180

pills detected on real phone photos (15 OTC products)

1.25%

false-accept rate, 9 of 720 wrong-profile trials (1.15% held-out CV)

31.1%

first-shot verify — deliberately low; the system prefers to abstain

1 → 0

safety events, shipped Rx parser (24 labels): the one recorded event was closed by a label-side guard (G6)

81%

answers surfaced in top-5 passages after intent routing (was 70%)

\$0.49

cloud cost for the full 120-question Q&A evaluation (~¼ each)

The finding we didn't expect: zero-shot models beat our fine-tuned ones on 2 of 3 vision heads on real photos. Studio benchmarks overstated real-world transfer — a negative result that now shapes the pipeline and anchors the paper.

All figures are development-set diagnostics with pre-registered pass/fail bars — quoted as evidence of method, never as product claims. Confirmatory capture study: pending (right panel).

PillSafeTray v2

WORK IN PROGRESS



Design render — not a photograph. Tray body is print-ready; the confirmatory capture study is pre-registered and pending.

- Bank-card footprint (**85.6 × 54 mm**); six wells, imprint-up; doses of 7–8 use two loads.
- Calibrated colour patches + four corner markers — the phone photo becomes a small, repeatable studio shot.
- Flash forced on: measured to rescue a dim room to ≈ daylight verify rates — lighting is the dominant capture lever measured (12/29 first-shot proxy; hard debossed imprints fail most).
- Pending:** 88-photo pilot → pre-registered 600-photo confirmatory; endpoints M1 usability, M2 safety (0 false accepts over ~630 strays ⇒ ≤0.48% by rule of three).

Scientific Foundation

- NLM Pill Image Recognition Challenge** — IEEE AIPR 2016 · the 43% that reframed the task
- MobileDeepPill** — MobiSys 2017 · attribute decomposition on-phone
- ePillID** — CVPR-W 2020 · the low-shot reality of pill imagery
- GO-PILL** — MDPI Mathematics 2026 · debossed imprints are the hard OCR case
- MEDIC** — Nature Medicine 2024 · guarded, catalogue-checked Rx extraction beats raw LLMs; our six guardrails descend from its five
- Hanley & Lippman-Hand** — JAMA 1983 · the rule of three sizes our safety claims

In preparation: "Verify, Don't Identify: Profile-Constrained Pill Verification from Consumer Photos with a Printed Capture Card" — the negative results are contributions.

Future Work

- Print the tray + run the pre-registered capture study (NB08, incl. multi-pill scenes).
- The imprint decision: stay zero-shot, or fine-tune on independent hand-keyed labels.
- Colour-boundary refinement (white/grey split first) · cheque-deposit-style auto-capture.

Team

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